

The World Might Not Be as Complex as We Thought

The Roles of Scale, Perspective, and Information in Human Understanding

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Abstract

A phenomenon can appear fragmented or difficult to understand because the conditions under which we encounter it reveal only part of its structure, or because we have yet to connect the information already available. We examine this possibility through a framework that distinguishes two analytical stages: a world, object, or process becomes available as information under a particular scale and perspective, and an analyst organizes that information into understanding. This distinction helps locate different sources of apparent complexity. A relevant relationship may remain inaccessible under the operative observation conditions; its informational basis may be available but insufficiently organized; or descriptions obtained under different conditions may each be intelligible while their connection remains unclear. Physical, organizational, temporal, and mathematical examples develop these possibilities and show why additional observations and better understanding need not advance together. An aggregation identity provides an explicit example of a bridge between descriptions at different scales. We then situate the argument in earlier work on observation, scale, information, explanation, and connections among theories. The resulting account offers a way to investigate where an apparent difficulty arises and what kind of inquiry may address it. The world can remain rich and genuinely difficult while some of its apparent complexity reflects how its relationships become accessible and connected to us.

Keywords: apparent complexity; scale; perspective; information; human understanding; relational fragmentation; cross-scale relations

1. Apparent Complexity

A three-dimensional cuboid can present different visible shapes as an observer moves around it. Viewed directly from one side, it may appear to be a rectangle; viewed from an angle, it may expose several faces and suggest depth. The object remains the same, but what the observer can learn from a particular view changes. If the observer has only one view, a simple object can support an incomplete account. If several views are available but their connection is unclear, the same object can support several individually intelligible yet apparently inconsistent accounts.

This example raises a broader question: why can a phenomenon appear fragmented, difficult, or discontinuous even when a more connected understanding of it is possible? A measurement can miss a relationship that lies outside its useful range. A collection of records can contain the information needed for an explanation without making that explanation apparent. Descriptions developed at different scales can each make sense while their relationship remains unresolved. These difficulties call for different responses, even though all can make a phenomenon appear complex.

We use *apparent complexity* to describe an analyst's encounter with a phenomenon as fragmented, irregular, disconnected, or discontinuous. The term identifies a difficulty in the account available to the analyst. Whether that difficulty reflects the phenomenon's structure, the conditions of access, or the organization of information requires further investigation. An accurate local description can still leave a larger relationship unexplained; an accurate projection can still conceal a dimension important to the question being asked.

We examine these possibilities by distinguishing two stages in the formation of understanding. First, a phenomenon becomes available as information under particular conditions. Second, an analyst organizes that information into an intelligible account. A difficulty in the first stage may require different observation conditions. A difficulty in the second may require better use of information already available. When descriptions obtained under different conditions remain disconnected, the inquiry must also establish how those descriptions relate.

The discussion follows this progression. We introduce the two-stage distinction, examine the difficulties that can arise within each stage, and then develop the roles of scale and perspective through examples. Those examples prepare the question of how separately intelligible accounts can be connected. We conclude by situating the argument in prior work and considering how the distinction can guide inquiry when a phenomenon continues to appear complex.

2. A Framework of World Revelation and Human Understanding

2.1 From the world to information

Consider a world, object, or process of interest, denoted by W . We encounter it at some scale, σ , and from some perspective, π . Scale describes the level or range at which we access the phenomenon. Depending on the question, it may refer to physical size, measurement range, duration, or aggregation across observations. Perspective describes the position or orientation from which the accessible phenomenon is encountered. In the cuboid example, the viewing angle changes perspective while the object and the level of description remain approximately fixed.

Let I denote the information about W made available under these conditions. We represent the relationship as:

$$I = \mathcal{R}(W; \sigma, \pi). \quad (1)$$

Equation (1) names the process through which the phenomenon becomes available as information. The revelation mapping, $\mathcal{R}$, is schematic: its particular operation depends on the application. A physical instrument, a set of organizational records, and a visual projection provide information in different ways. What they share is conditioned access to the phenomenon. The scale and perspective matter because they help determine what the resulting information can support.

The distinction between scale and perspective gives us two questions to ask about that access. Are we observing at a level or range suited to the relationship of interest? Are we encountering the phenomenon from a position that exposes the relevant features? A household scale and an instrument capable of

resolving atomic masses differ in measurement range. Two observers viewing a cuboid from different angles differ in perspective. Either difference can change the available information while the phenomenon of interest remains the same.

2.2 From information to understanding

Having information available creates the possibility of understanding, but realizing that possibility requires further work. An analyst may possess observations, statements, measurements, or records and still miss the relation that makes them jointly intelligible. Let A denote the analyst and U the understanding formed from the available information. We write:

$$U = \mathcal{G}_A(I). \quad (2)$$

In Equation (2), $\mathcal{G}_A$ represents the analyst's organization and interpretation of the information. The subscript recognizes that the resulting account depends on the analyst's knowledge, concepts, and analytical practices as well as on the material being examined. For a particular application, we treat that background as part of the operative analyst. Changes in the analyst's background or methods can therefore change the understanding supported by the same observations.

Combining the two stages gives:

$$U = \mathcal{G}_A(\mathcal{R}(W; \sigma, \pi)). \quad (3)$$

Equation (3) makes the dependency explicit. Understanding depends both on what becomes available and on how the analyst organizes it. The notation supplies a compact language for locating difficulties along this path. We use it as a conceptual representation rather than a quantitative measure of complexity.

The two stages are analytical distinctions within an inquiry that can proceed iteratively. An emerging understanding can lead an analyst to choose another instrument, examine a longer period, or seek another perspective. That choice changes the information available for the next round of analysis. The separation remains useful because it distinguishes improving access from improving the organization of what has already been accessed.

Table 1 summarizes the notation before we examine these two kinds of difficulty in turn.

Table 1. Key Elements and Notation of the Framework

Symbol	Meaning	Plain-language reading
W	World, object, or process of interest	What is being encountered or studied
σ	Scale	The level, range, duration, or aggregation at which the phenomenon is accessed
π	Perspective	The position or orientation from which the phenomenon is encountered
I	Information	What becomes available under the operative conditions
A	Analyst	The person or analytical process organizing and interpreting the information
U	Understanding	The account formed from the available information
$\mathcal{R}$	Revelation mapping	How the phenomenon becomes available as information
$\mathcal{G}_A$	Understanding mapping	How analyst A organizes and interprets the information

3. Conditioned Revelation

The first difficulty arises when the observation conditions fail to expose a relationship relevant to the inquiry. In the notation above, the problem lies in the transition from W to I . The analyst may have reason to expect a relationship, yet the information supplied under the current conditions may provide little or no basis for examining it.

Consider the relationship between the number of hydrogen atoms and their total mass. Under comparable physical conditions, adding atoms adds mass. An ordinary household scale, however, cannot resolve the difference between one atom and a few atoms. The relationship can be established through other means, but the readings from this instrument do not provide the evidence needed to examine it at that range.

This example helps distinguish two reasons a relationship may remain difficult to observe. Sometimes a suitable measurement process produces noisy or sparse observations. Additional measurements can then improve an estimate or make an existing pattern easier to distinguish. In other cases, the recording process fails to preserve the relevant differences. If every atom count being compared produces the same recorded value and no count-dependent information survives in the record, repetition under those conditions cannot distinguish the counts. The useful response is to change the measurement conditions.

The qualification matters because a failure to see a relationship has several possible explanations. The relationship may be absent; it may be weak relative to noise; or the observation process may be unable to reveal it. An absence in the available record, by itself, leaves these possibilities unresolved. Establishing which explanation applies requires evidence about both the phenomenon and the means through which it is observed.

The cuboid presents a parallel case. A fixed two-dimensional view can preserve the outline of a face while concealing depth. Repeatedly inspecting that outline can improve knowledge of the visible shape, but depth may require another view or other information. In both examples, the question is what the operative conditions allow the analyst to learn.

Conditioned revelation therefore directs attention to the fit between the question and the conditions of access. When the relevant distinctions are not preserved, additional information of the same kind may leave the difficulty intact. Section 5 extends this point: changing scale can also change the form of the relationship that becomes useful. Before considering that extension, we turn to a difficulty that can persist even when the necessary information is already available.

4. Relational Fragmentation and Connected Understanding

Suppose an analyst already has the observations or records needed to examine a relationship. The remaining task is to recognize how they fit together. This places the difficulty in the transition from *I* to *U*: the informational basis is available, but the analyst's account has yet to incorporate the relevant connection.

A paragraph provides a familiar example. A reader may understand each sentence while missing how one sentence qualifies another, why an example appears at a particular point, or how the conclusion follows from the preceding discussion. The sentences can be individually clear while the argument remains obscure. Reading them as isolated statements will then reproduce the difficulty. Understanding the paragraph requires recognizing their roles in a developing argument.

An illustrative organizational case shows why the same distinction matters outside language. Suppose an analyst has records showing that a purchasing unit places large orders to reduce unit costs, a warehouse limits incoming deliveries because storage capacity is constrained, and a finance unit delays payments to preserve cash. Each action can appear reasonable when examined against its immediate objective. Taken together, the actions may look inconsistent: orders are placed, deliveries are restricted, and payments are delayed.

In this example, the objectives and constraints are already in the analyst's records. Connecting them reveals how one unit's action changes the conditions faced by another. The purchasing decision increases the warehouse's capacity burden, and delayed payments can constrain the implementation of the purchasing plan. The joint pattern becomes more intelligible when the analyst organizes the records around these dependencies. Understanding the pattern explains how the conflict arises; resolving the actual conflict is a further organizational task.

The relevant information need not state the relationship explicitly. It may instead provide the basis from which the analyst can recognize or derive the connection. In this sense, having the informational basis available is different from already understanding what it implies.

Holding the available records fixed makes this a case of organization rather than expanded access. If the analyst lacks the information about storage capacity or cash constraints, the inquiry must also obtain that

information. A practical problem can thus involve both stages. The distinction helps identify the work still required without forcing every difficulty into a single category.

We use *relational fragmentation* for the condition in which relevant informational elements remain insufficiently connected in the analyst's understanding. The connections may involve sequence, dependence, comparison, qualification, or aggregation. Their appropriate form depends on the question. The paragraph requires an argumentative relationship; the organizational case requires relationships among objectives, constraints, and actions.

This explains why accumulating information can sometimes increase apparent complexity. Additional reports or specialist descriptions may enlarge what the analyst knows separately while increasing the number of relationships still to be understood. More records help when they supply a missing basis for explanation. When that basis is already available, progress depends on organizing it into a connected account. We can now examine how scale and perspective affect the information to be organized, including cases in which a different condition makes a different relationship relevant.

5. Scale and Perspective in Apparent Complexity

5.1 Scale: different levels can reveal different relationships

Scale affects more than the precision with which we observe a fixed relationship. It can also change the relational description needed to understand the observations. Aggregation makes this distinction explicit because a relationship among individual values can take a different form when expressed in terms of their averages.

Consider n paired observations, with n a positive integer. Let x_i and y_i be dimensionless numerical values for observation i , where $i = 1, \dots, n$. Suppose each pair satisfies:

$$y_i = x_i^2. \quad (4)$$

Equation (4) supplies a complete relationship for each individual pair. To examine the same observations at an aggregate level, define their means and the variance of the x_i values as follows:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i, \quad v_x = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2. \quad (5)$$

Here variance measures how far the observed values vary around their mean. Expanding the squared deviations gives:

$$v_x = \frac{1}{n} \sum_{i=1}^n x_i^2 - 2\bar{x}^2 + \bar{x}^2 = \bar{y} - \bar{x}^2. \quad (6)$$

Rearranging Equation (6) gives $\bar{y} = \bar{x}^2 + v_x$. This identity shows why the average relationship differs from the individual relationship. Although each y_i equals the square of its corresponding x_i , the average of the y_i values also depends on dispersion among the x_i values. The square of the average alone leaves out that contribution.

For a numerical check, take $x_1 = 1$ and $x_2 = 3$. Their corresponding y_i values are 1 and 9. The means are therefore $\bar{x} = 2$ and $\bar{y} = 5$, while $v_x = 1$. The aggregate identity gives $5 = 2^2 + 1$. If all the x_i values are

equal, the variance is zero and the aggregate relationship reduces to $\bar{y} = \bar{x}^2$. Thus the additional term arises precisely from variation across the observations.

The same identity also holds in probability when the relevant expectations exist. In either form, variance describes variation across a collection or distribution, rather than a separate value attached to an individual observation.

Temporal scale can also change what an account can explain. Suppose a delivery is late at the time it is inspected. A snapshot records its status. A longer observation period may show when the order was approved, when dispatch occurred, and where transportation was delayed. Those records expose a sequence through which the observed status developed. The sequence supports questions about possible mechanisms that a snapshot leaves inaccessible. Establishing a causal explanation requires further evidence about the relevant dependencies; temporal order supplies part of the basis for that inquiry.

A similar possibility arises across spatial levels. A driver can encounter irregular braking and merging, while information about a bottleneck and the propagation of queues can make part of that local pattern intelligible. This illustrative case allows local difficulty and broader intelligibility to coexist. The broader account helps when it connects the local observations to relevant system relationships. Its value comes from that connection, rather than from breadth alone.

5.2 Perspective: different vantage points can reveal different projections

The preceding examples change the level or range of description. Perspective concerns the position from which an accessible phenomenon is encountered. Returning to the cuboid, hold its length, width, and height fixed and change the viewing angle. A face visible from one position may be concealed from another; the same dimensions can produce different projected outlines.

For this geometric case, let P_π denote the projection associated with viewing perspective π . We can write:

$$I_\pi = P_\pi(W). \quad (7)$$

Equation (7) illustrates conditioned revelation with a specific operation: projection. Different projections can accurately represent different aspects of the same object. Their compatibility becomes easier to recognize when the observer knows the viewing positions and the geometry connecting them.

Perspective can matter in organizational inquiry as well. An employee, a manager, and an external observer may encounter different interactions and constraints because their positions provide different access. Each can develop a locally useful account of the same process. Comparing those accounts requires attention to what each position exposes and what it conceals. Position-dependent differences are therefore a reason to investigate access before treating differences in description as disagreements about the underlying process.

Scale and perspective often change together. Examining the organizational case at the level of several units expands the range of the inquiry, while obtaining access to a manager's records changes the position from which dependencies become visible. Keeping the concepts distinct helps explain what each change contributes, even when both occur in the same investigation.

These examples establish why changing observation conditions can provide different relationships as well as additional observations. They also create the next question. If different conditions support different intelligible accounts, what allows the analyst to understand those accounts together?

6. Across Conditions: Connection, Transition, and Apparent Discontinuity

Consider information obtained from the same phenomenon under two sets of conditions. Using subscripts to distinguish those conditions, we write:

$$I_1 = \mathcal{R}(W; \sigma_1, \pi_1), \quad I_2 = \mathcal{R}(W; \sigma_2, \pi_2). \quad (8)$$

For the operative analyst A , let the resulting understandings be $U_1 = \mathcal{G}_A(I_1)$ and $U_2 = \mathcal{G}_A(I_2)$. Each account may be intelligible on its own. The remaining task is to establish the relationship between them. Knowing what each account says can leave unresolved why their descriptions differ or how one relates to the other.

We use *bridge*, denoted by B_{12} , for knowledge that explains this relationship. The subscript identifies the two accounts being connected. A bridge must supply more than a statement that both concern the same phenomenon. It must explain a relevant connection, such as how aggregation changes a relation, how a projection preserves some features and conceals others, or how a process carries a system from one observed state to another.

The aggregation example provides an explicit bridge. At the individual level, $y_i = x_i^2$. At the aggregate level, $\bar{y} = \bar{x}^2 + v_x$. The derivation in Equation (6) establishes how the second description follows from the first and why the variance term appears. An analyst who expects the aggregate relation to be $\bar{y} = \bar{x}^2$ can now identify the source of the difference. The bridge connects the descriptions by explaining what averaging does to a nonlinear relationship.

The cuboid provides a different kind of bridge. Viewing positions and three-dimensional geometry explain how different two-dimensional outlines can arise from one object. Knowing the outline alone does not generally recover the object's full geometry. The connection can therefore be informative without making the descriptions interchangeable or the projection reversible. This is why the particular form of a bridge must be established for the application rather than assumed from the fact that two accounts exist.

We use *apparent discontinuity* for the analyst's experience of accounts as disconnected when their relationship remains unclear. Individually coherent descriptions can generate this experience if the connection between them has yet to be explained. The gap then identifies a question for inquiry: what would establish whether these descriptions are compatible, and how are they related if they are?

Coherence alone cannot answer that question. An account can be internally coherent and still be inaccurate, or two accounts may concern conditions that support a genuine change in the phenomenon. A proposed bridge must therefore be supported by the relevant evidence or derivation. The aggregation identity meets that requirement mathematically. A geometric explanation of the cuboid must fit the actual

object and viewing conditions. In other applications, examining compatibility may lead to revising an account rather than connecting the accounts as initially stated.

Changing observation conditions can help this inquiry by providing evidence about the connection itself. Following a change in viewing angle can show how one outline develops into another. Examining intervening records can clarify a temporal sequence. Retaining the individual observations while computing their means can reveal the role of dispersion. These changes are useful because they make the relationship between descriptions examinable.

The connection problem thus extends the two-stage distinction. We may need better access to the phenomenon, better organization of available information, or knowledge that relates accounts produced under different conditions. These possibilities have recognizable counterparts in earlier work. We turn to that work to clarify the intellectual setting of the argument.

7. The Framework in Relation to Prior Work

7.1 Conditioned access to the world

The transition from *W* to *I* connects to work on observation and representation. Hanson examined how conceptual commitments shape observation, and Kuhn situated scientific perception within the paradigms through which inquiry proceeds [1,2]. Giere's scientific perspectivism addresses the selective representation of the world through models, instruments, and observational positions [3]. Massimi's perspectival realism develops the possibility of world-directed knowledge across scientific perspectives [4]. Together, these works help situate the idea that access is conditioned while inquiry can still seek knowledge of the phenomenon.

Our immediate question concerns how that conditioning affects apparent complexity. The cuboid makes selectivity visible in a literal projection; the household-scale example makes it visible in measurement range. These examples allow us to examine a particular difficulty: the available information can leave out a relationship needed for the question being asked. Earlier work provides a broader intellectual setting for treating the conditions of observation as part of the analysis.

7.2 Scale and the form of a useful description

Work on levels and scale helps situate the distinction between improving resolution and changing the form of a relation. Oppenheim and Putnam considered levels of scientific description, Simon examined hierarchical organization, and Anderson emphasized that higher-level behavior can require concepts beyond those used in lower-level descriptions [5–7]. Wilson's work on renormalization, Batterman's examination of asymptotic reasoning, and Noid's discussion of coarse-grained modeling provide different settings in which scale and effective description are closely related [8–10].

The aggregation example gives this discussion a limited, explicit case. A relation among averages requires a term describing variation across individuals, and the derivation shows why. That result supplies the connection needed here without deciding broader questions about emergence or reduction. The earlier

literature helps readers place the example within the larger problem of how descriptions at different levels are related.

7.3 Information organized into explanation

The second transition, from *I* to *U*, concerns the use of available information. Shannon's mathematical theory defines a precise quantitative concept of information for communication systems [11]. Here, information has the broader meaning of observations, records, statements, and other material available to an analyst. This distinction is important because our question concerns what that material supports when organized into an account, rather than how much information it contains under a communication-theoretic measure.

Hutchins examined cognitive accomplishment through the organization of information across people, representations, and artifacts [12]. Bechtel and Richardson, Craver, and Wimsatt address decomposition, organization, and relations among parts in scientific explanation [13–15]. These works help situate the point illustrated by the paragraph and organizational examples: understanding depends on the connections that make separate elements jointly intelligible. The form of those connections varies with the problem, but their explanatory role gives organization a place distinct from accumulation.

7.4 Connections among accounts

The bridge problem connects to earlier work on relations among theories. Nagel's treatment of intertheoretic reduction and Schaffner's later analysis examine ways to connect descriptions expressed in different theoretical vocabularies [16,17]. This tradition makes clear why having descriptions on both sides of a divide leaves further analytical work to be done.

We use a bridge more broadly for the knowledge that explains a relevant relationship between conditioned accounts. An aggregation identity, a geometric projection, and a temporal sequence can each supply a different kind of connection. Their value must be assessed against the relationship they claim to explain. Earlier work helps locate this task historically; the present argument organizes it around a practical concern shared with the two preceding difficulties: why a phenomenon appears disconnected to an analyst and what would make its relationships more intelligible.

8. Discussion and Conclusions

The starting question was whether some of the difficulty we encounter reflects how we access and organize a phenomenon. Distinguishing conditioned revelation from the formation of understanding makes that question more specific. It directs inquiry toward the information available, the relationships recognized within it, and the connections established among accounts obtained under different conditions.

When a relationship cannot be examined through the available record, the analyst can investigate the conditions of access. The atom-count example illustrates why a suitable measurement range matters. The cuboid illustrates why a different viewpoint can expose a concealed dimension. In both cases, progress depends on information capable of supporting the question, and evaluating the observation process helps identify what kind of change may supply it.

When the necessary informational basis is already available, the analyst can examine its organization. In the paragraph example, the task is to connect the sentences as parts of an argument. In the organizational example, the task is to connect actions through objectives, constraints, and dependencies. These connections can make a pattern understandable even when the underlying practical conflict remains. Explanation and resolution answer related but different questions.

When separately intelligible accounts remain difficult to reconcile, the analyst can investigate their relationship. The variance identity shows how a bridge can explain a change in descriptive form exactly. The cuboid shows how accurate descriptions can differ through projection. Other cases require evidence to determine whether the accounts are compatible, whether one needs revision, or whether a genuine transition in the phenomenon accounts for the difference.

These tasks can interact within one inquiry. A missing organizational record can require expanded access before the analyst can construct the relevant explanation. An emerging explanation can suggest which record to seek next. A proposed connection between accounts can identify observations needed to test it. The framework helps direct that work by asking where the current account is incomplete and what would address that incompleteness.

Its value lies in the fit between a difficulty and the next analytical step. Broader coverage may expose a dependency while also adding distinctions to be explained. A different perspective may resolve one ambiguity and introduce another. Additional information can help when it supplies a missing basis for explanation; better organization can help when that basis is already available. Each response should be judged by the relationship it enables the analyst to examine.

The title follows from this possibility. The world can remain rich, multidimensional, and genuinely difficult while some apparent disorder reflects a limited view, some apparent inconsistency reflects disconnected local accounts, and some apparent discontinuity reflects a relationship yet to be established. Recognizing these possibilities gives us a more precise way to investigate difficulty. In that sense, the world might not be as complex as we thought: understanding can advance when we improve the conditions under which its relationships become visible and the connections through which they become intelligible.

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